Jacob Pieniazek

Lead Data Scientist | Causal Inference & Technical Leadership

jacob-pieniazek.com | jakepenzak | japieniazek | jacob@pieniazek.me | +1 (443) 994-2344

Experience

84.51° (KROGER) | Lead Data Scientist

Feb 2026 – Present | Cincinnati, OH

- ➔ Set scientific direction and lead technical delivery for five data scientists measuring 1,000+ marketing campaigns annually in a production experimentation platform, informing dozens of Kroger Marketing stakeholders.
- ➔ Lead a two-scientist causal ML R&D team whose targeting consistently delivers roughly 2x the causal lift of business-as-usual targeting across deployed production use cases; translate non-linear CATE patterns into marketing insights.
- ➔ Negotiated, designed, and led a 100K-household randomized audit of a partner paid on incremental profit. Found partner-reported lift was roughly twice the experimental estimate and recalibrated profit-sharing payments, preventing significant overpayment.
- ➔ Deliver millions in annual cloud/platform savings by modernizing infrastructure, right-sizing Databricks compute, and optimizing PySpark pipeline execution.
- ➔ Established Claude Code-assisted development practices and AI-generated CI summaries for the development team.
- ➔ Develop LangChain-based agentic pipelines for econometric reporting and insights.
- ➔ Mentor four new hires and lead Miami University data science recruiting. Co-lead a biweekly causal inference learning group. Designed interview cases and rubrics and co-built internal causal inference training for scientists and business teams.

84.51° | Senior Data Scientist

Jul 2024 – Feb 2026 | Cincinnati, OH

- ➔ Architected a Python measurement platform using Kedro and shared Ibis transformations for interoperability across local DuckDB and production PySpark. Migrated Azure Data Factory workflows to GitHub Actions and Databricks, cutting pipeline costs by 50%.
- ➔ Measured roughly $120M in full-year FY2024 incremental sales for a personalization program spanning 30+ campaigns and 20M households. Designed and deployed its experimental framework with sticky household treatment assignments and a common allocation ratio across entry cohorts to capture cumulative program effects.
- ➔ Integrated Double Machine Learning & modern staggered Difference-in-Differences estimators into measurement, with randomized placebo tests to check for spurious lift.

84.51° | Data Scientist

Jun 2022 – Jul 2024 | Cincinnati, OH

- ➔ Corrected email-suppression bias using offer-specific intent-to-treat or instrumental-variable estimators and added automated design diagnostic checks to production campaign measurement.
- ➔ Integrated causal, experimental, and quasi-experimental econometric methods into measurement workflows and served as a key proponent for wider adoption of causal reasoning & methods across the organization.

Open Source & Personal Projects

CAML | Causal ML & AutoML (docs) (code)

- ➔ Building an open-source Python framework to automate selection and validation of heterogeneous treatment-effect models using EconML, FLAML, and Optuna.

NETBEAT | Network Diagnostics CLI (crates) (code)

- ➔ Built a Rust client/server CLI for throughput and latency diagnostics, with concurrent TCP connections and JSON reports for automation.

SELF-HOSTED AI PLATFORM | Personal LLM Infrastructure

- ➔ Build and operate a personal AI platform with centralized cloud/local model routing (LiteLLM), GPU-backed inference, managed multi-user access, agent workflows, persistent memory, and semantic retrieval.

Skills

CORE STRENGTHS

- Technical Leadership
- Causal ML & Experimentation
- Applied Econometrics
- Data Engineering
- MLOps, DevOps, CI/CD
- Production & AI Systems

PROGRAMMING

Python • SQL • Rust
Bash • R • Stata

ML / TOOLS

scikit-learn • PyTorch • XGBoost
EconML • DoubleML • CausalML
FLAML • Optuna • MLflow

AI / AGENTS

LangChain • MCP, A2A
Agent Harness (eg Claude Code)
LiteLLM • Embeddings
Agentic Development

SYSTEMS / MLOPS

Databricks • PySpark • Delta
Kedro • Ibis • DuckDB
Azure • Terraform
GitHub Actions • FastAPI
Linux • Docker • Kubernetes
PostgreSQL • pytest

Education

MIAMI UNIVERSITY

M.A. Economics | GPA: 4.0
2022 | Full Graduate Assistantship

UNIVERSITY OF DAYTON

B.S. Business Economics & Applied Mathematical Economics
Double Major | 2021 | GPA: 3.92
Summa Cum Laude, Honors Diploma
E. B. O'Leary Award of Excellence

INVITED TALKS

Causal Machine Learning in Practice
Miami University Economics Brown Bag 2026

Interests/Hobbies

Open-Source Development • Self-Hosting / Homelab • Home Automation • Hiking / Camping • Guitar • Audiophile Pursuits